11. Synthetic Medical Test Reports

The following 5 synthetic patient reports are designed for end-to-end testing of MediAgent. Each report is based on realistic clinical scenarios and structured to trigger specific system behaviors. These can be typed directly into the symptom textarea or used as reference for building synthetic PDF lab reports.

⚠ Disclaimer

- All patient data below is entirely fictional and fabricated for software testing purposes.
- Any resemblance to real persons is coincidental.
- These reports must NOT be used for actual medical decision-making.

📄 SYNTHETIC TEST REPORT — REPORT A — Community-Acquired Pneumonia	
Patient Name	Ravi Sharma (SYNTHETIC)
Age / Gender	52 years / Male
Date	April 15, 2026
Chief Complaint	High-grade fever, productive cough, chest pain — 5 days duration
Full Symptom Description	52-year-old male presenting with 5 days of high-grade fever reaching 39.8°C, productive cough with yellow-green sputum, right-sided pleuritic chest pain worsening on deep inspiration. SpO2 93% on room air at rest. Patient is a non-smoker with no prior lung disease. Lives in a densely populated urban area. Recent sick contacts at workplace.
Lab — Hemoglobin	11.8 g/dL (LOW — ref 13.5-17.5)
Lab — WBC	18,400/µL (HIGH — ref 4,000-11,000)
Lab — Neutrophils	85% (HIGH — ref 40-70%)
Lab — CRP	142 mg/L (HIGH — ref < 5 mg/L)
Lab — Procalcitonin	1.8 ng/mL (HIGH — ref < 0.5 ng/mL)
Lab — Serum Sodium	131 mEq/L (LOW — ref 136-145)
Lab — Serum Albumin	2.9 g/dL (LOW — ref 3.5-5.0)
Vision Context	Chest X-ray (if uploaded): Expected findings — right lower lobe infiltrate, air bronchogram sign
Image Type	chest_xray (expected if actual CXR uploaded)
Severity Hint	moderate
Expected Top Diagnosis	Community-Acquired Pneumonia (J18.9) — Likelihood: HIGH
Expected Secondary	COVID-19 Pneumonitis (U07.1) — Likelihood: MODERATE

Expected RAG Patterns	Bilateral or unilateral pulmonary involvement with systemic inflammatory signs
Expected Red Flags	SpO2 93% — possible respiratory compromise; high WBC + CRP = sepsis risk
Expected Urgency	urgent or immediate
Expected Next Steps	Blood cultures x2 before antibiotics; CBC with differential; sputum gram stain; chest X-ray; COVID-19 PCR; pulse oximetry monitoring
Test Use	Enter the Symptom Description into the MediAgent symptom textarea. Upload a HAM10000 image to test vision pipeline (it will misclassify as skin lesion — that is expected).

📋 SYNTHETIC TEST REPORT — REPORT B — Diabetic Ketoacidosis (DKA)

Patient Name	Priya Menon (SYNTHETIC)
Age / Gender	28 years / Female
Date	April 10, 2026
Chief Complaint	Vomiting, confusion, abdominal pain — 1 day duration
Full Symptom Description	28-year-old female with Type 1 Diabetes Mellitus (diagnosed age 14) presenting with 18 hours of nausea, vomiting, diffuse abdominal pain, and progressive confusion. Her partner reports she has had 'fruity-smelling' breath. She was unable to eat or drink fluids for 12 hours. Heart rate 118 bpm. Blood pressure 100/60 mmHg. Respiratory rate 28 breaths/min with deep, labored breathing (Kussmaul respiration). Random glucose > 400 mg/dL.
Lab — Blood Glucose	487 mg/dL (HIGH — ref 70-100 fasting)
Lab — Arterial pH	7.18 (LOW — ref 7.35-7.45)
Lab — Bicarbonate	8 mEq/L (LOW — ref 22-28)
Lab — Serum Ketones	STRONGLY POSITIVE
Lab — Anion Gap	28 (HIGH — ref 8-12)
Lab — Serum Potassium	6.1 mEq/L (HIGH — ref 3.5-5.1)
Lab — Serum Sodium	128 mEq/L (LOW — ref 136-145)
Lab — Creatinine	1.9 mg/dL (HIGH — ref 0.5-1.1 female)
Lab — WBC	14,200/µL (HIGH — stress leukocytosis)
Expected Top Diagnosis	Diabetic Ketoacidosis (E10.10) — Likelihood: HIGH
Expected Secondary	Hyperosmolar Hyperglycemic State (E11.01) — Likelihood: LOW
Expected Red Flags	Altered mental status + sepsis-like picture; Kussmaul respiration (respiratory distress terminology); neurological deficit (confusion)
Expected Urgency	immediate

Expected Next Steps	ICU-level monitoring; IV insulin infusion protocol; IV fluid resuscitation (0.9% saline); hourly electrolytes; cardiac monitoring for hyperkalemia
Test Use	Use this symptom text to test that altered mental status + respiratory distress terms trigger the immediate urgency escalation path in the Report Agent.

🗒️ SYNTHETIC TEST REPORT — REPORT C — Pulmonary Embolism (Suspected)

Patient Name	Ahmed Al-Farsi (SYNTHETIC)
Age / Gender	44 years / Male
Date	April 12, 2026
Chief Complaint	Sudden-onset dyspnea and pleuritic chest pain — 6 hours
Full Symptom Description	44-year-old male, 10 days post right knee arthroscopy, presenting with sudden-onset dyspnea at rest and right-sided pleuritic chest pain. He reports mild hemoptysis (blood-streaked sputum) twice in the last 4 hours. Right calf is swollen, warm, and tender. Heart rate 110 bpm, SpO2 91% on room air. Patient flew long-haul (9-hour flight) 2 weeks ago. Wells score estimated at 7.5 (high probability).
Lab — D-Dimer	4,800 ng/mL (HIGH — ref < 500 ng/mL)
Lab — Troponin I	0.08 ng/mL (borderline HIGH — ref < 0.04)
Lab — NT-proBNP	620 pg/mL (HIGH — ref < 125)
Lab — WBC	11,800/µL (high normal)
Lab — ABG pO2	68 mmHg (LOW — ref 80-100 on room air)
Lab — ABG pCO2	32 mmHg (LOW — ref 35-45, hyperventilation)
Vision Context	CT Pulmonary Angiography (if available) — Expected: filling defect in right main pulmonary artery
Image Type	ct_abdomen or unknown (LLaVA may misclassify without actual CT)
Severity Hint	severe
Expected Top Diagnosis	Pulmonary Embolism (I26.99) — Likelihood: HIGH
Expected Secondary	Pneumonia with pleurisy (J18.9) — Likelihood: LOW
Expected Red Flags	SpO2 91% + hemoptysis + respiratory distress = at least 2 red flags; cardiac compromise (elevated troponin)
Expected Urgency	immediate
Test Use	Critical test for the multi-factor red flag path. SpO2 91% + respiratory distress + hemoptysis should trigger 2+ red flags → 'immediate' urgency. Verify the urgency logic precisely.

📄 SYNTHETIC TEST REPORT — REPORT D — Routine Lipid Panel (No Red Flags)

Patient Name	Sunita Patel (SYNTHETIC)
Age / Gender	38 years / Female
Date	April 14, 2026
Chief Complaint	Annual health check — no acute symptoms
Full Symptom Description	38-year-old female, non-smoker, no significant past medical history, presenting for annual preventive health check. No symptoms of chest pain, dyspnea, or palpitations. Family history of Type 2 Diabetes in father. BMI 26.4 kg/m². Blood pressure 118/76 mmHg. Regular exercise 3x/week. Occasional social alcohol.
Lab — Total Cholesterol	228 mg/dL (BORDERLINE HIGH — ref < 200)
Lab — LDL Cholesterol	148 mg/dL (HIGH — ref < 100 optimal)
Lab — HDL Cholesterol	52 mg/dL (NORMAL — ref > 50 female)
Lab — Triglycerides	182 mg/dL (BORDERLINE HIGH — ref < 150)
Lab — Fasting Glucose	98 mg/dL (NORMAL — ref 70-99)
Lab — HbA1c	5.6% (NORMAL — ref < 5.7%)
Lab — TSH	2.1 mIU/L (NORMAL — ref 0.4-4.0)
Lab — Hemoglobin	13.2 g/dL (NORMAL)
Expected Top Diagnosis	Hyperlipidemia (E78.5) — Likelihood: HIGH
Expected Secondary	Pre-diabetes risk monitoring — Likelihood: LOW
Expected Red Flags	NONE — no acute symptoms, no high-risk terms
Expected Urgency	routine
Expected Next Steps	Lifestyle modification counseling (diet, exercise); consider statin therapy per ATP III guidelines; repeat lipid panel in 6 months; annual HbA1c given family history
Test Use	Verify that the system correctly assigns routine urgency with zero red flags for a non-acute preventive case. This tests the lower bound of the urgency logic.

📄 SYNTHETIC TEST REPORT — REPORT E — Skin Lesion Dermoscopy (HAM10000 Compatible)

Patient Name	Arjun Nair (SYNTHETIC)
Age / Gender	55 years / Male
Date	April 16, 2026
Chief Complaint	Changing mole on left forearm — 3 months

Full Symptom Description	55-year-old male with Fitzpatrick skin type III, prolonged UV sun exposure (outdoor worker for 20 years), presenting with a changing pigmented lesion on the left forearm. The lesion has grown from approximately 4mm to 8mm over 3 months. ABCDE criteria assessment: A — Asymmetry: present; B — Border: irregular, scalloped; C — Color: variegated (dark brown, black, pink areas); D — Diameter: 8mm (> 6mm threshold); E — Evolution: rapid growth over 3 months. No regional lymphadenopathy. No systemic symptoms.
Dermatoscopy Finding	Atypical pigment network, irregular dots/globules, regression structures
Lesion Location	Left forearm, volar surface
Lesion Size	8mm × 7mm
ISIC Archive Reference	HAM10000 — Melanocytic Nevi (nv) / Melanoma (mel) category
Vision Context (HAM10000 image)	Upload any ISIC_*.jpg from manual-test/images/ham10000/
Expected image_type	skin_lesion (LLaVA should classify this correctly)
Expected Severity	moderate to severe (irregular features)
Expected Anomalies	Atypical pigment patterns, irregular border
Expected Top Diagnosis	Melanoma (C43.60) or Atypical Melanocytic Nevus — Likelihood: HIGH
Expected Secondary	Seborrheic Keratosis (L82.1) — Likelihood: LOW (lacks typical features)
Expected Red Flags	Possible for rapid evolution + ABCDE criteria trigger (if red-flag keywords detected in input)
Expected Urgency	urgent (rapid evolution + multiple ABCDE criteria)
Expected Next Steps	Urgent dermatology referral within 2 weeks; complete excisional biopsy with 1mm margins; full skin examination; sentinel lymph node biopsy if melanoma confirmed
Test Use	BEST TEST for Vision Agent. Upload a real HAM10000 image + this symptom text simultaneously. Verify LLaVA classifies the image as skin_lesion and the report integrates both image findings and symptom analysis.